GLUEING FINITELY MANY ONE-DIMENSIONAL SEMI-LOCAL RINGS INSIDE A FIELD INTO A SINGLE SEMI-LOCAL DOMAIN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Two-sided glueing of one-dimensional semi-local rings. Stacks Project, Varieties, Lemma lemma-semi-local-both-side (tag 09N0). Performs the induction step that combines one local ring with the intersection of the others.

Source reference. Original lemma 09N3: sources/stacks-chow/tags.txt, source lines 12201–12201. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let K be a field. Let $A_1, \ldots, A_r \subset K$ be Noetherian semi-local rings of dimension 1 with fraction field K. If $A_i \otimes A_j \to K$ is surjective for all $i \neq j$, then there exists a Noetherian semi-local domain $A \subset K$ of dimension 1 contained in $A_1, \ldots, A_r$ such that*

(1) *$A \to A_i$ induces an open immersion $j_i : \mathrm{Spec}(A_i) \to \mathrm{Spec}(A)$,*
(2) *$\mathrm{Spec}(A)$ is the union of the opens $j_i(\mathrm{Spec}(A_i))$,*
(3) *each closed point of $\mathrm{Spec}(A)$ lies in exactly one of these opens.*

Proof. Namely, we can take $A = A_1 \cap \ldots \cap A_r$. First we note that (3), once (1) and (2) have been proven, follows from the assumption that $A_i \otimes A_j \to K$ is surjective since if $\mathfrak{m} \in j_i(\mathrm{Spec}(A_i)) \cap j_j(\mathrm{Spec}(A_j))$, then $A_i \otimes A_j \to K$ ends up in $A_{\mathfrak{m}}$. To prove (1) and (2) we argue by induction on r. If $r > 1$ by induction we have the results (1) and (2) for $B = A_2 \cap \ldots \cap A_r$. Then we apply Lemma 09N3 to see they hold for $A = A_1 \cap B$. □

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